DYSMENORRHEA

Definition : It means painful menstruation "Pain may occur just before or during menstruation".

Incidence: 30% to 75%, average 50% of all females.

I- Dysmenorrhea without evident pelvic disease (primary – spasmodic).

***Definition:** It is colicky pain of uterine origin during menstruation with no pelvic lesion.

*** Causes:** the exact cause is not known but theories:

I- Abnormal uterine anatomy:

1- Uterine hypoplasia.
2- Uterine maldevelopment: e.g. bicornuate uterus.
3- Uterine malposition: e.g. retroverted flexed uterus.
4- Cervical stenosis.

II- Abnormal physiology:

1- Disturbed polarity of the uterus: the isthmus contracts during menstruation and not relaxed.
2- Functional or mechanical obstruction to the menstrual flow causes great ↑ of intra-uterine pressure to higher levels than the arterial blood pressure → ischemia of uterine muscles → local anoxia → stimulate nerve endings inside myometruim → angina like pain of uterus.

III- Prostaglandins PGs: ↑ local uterine PG especially $PGF_2\propto$ → ↑ uterine contraction.

IV- Psychological factors:

V- Allergy: to certain substances e.g. ↑ PG

C/P: 1- It occurs in virgin and nullipara.

2- It usually starts 2-3 years after puberty as the initial cycles usually is anovulatory and the primary dysmenorrhea occurs more with ovulatory cycles.

3- Pain starts with 1st day of menses.

4- Colicky suprapubic pain with radiation to back and inner and front of thighs.

5- Pain may be associated with vomiting and nausea (due to ↑ PG).

6- Pelvic examination reveals no abnormalities could be detected.

Treatment:

I- *General measure:*

- Education of young girls about physiology and hygiene of menstruation.
- Correction of anemia.
- If the woman is married, pregnancy is encouraged.

II- *Medical measures:*

1- Anti prostaglandins: e.g. aspirin or mefenamic acid.

2-Antispasmodics:
- Atropine
- Tocolytic drugs: B-agonist
- Calcium channel blockers.

3-Antihistaminics.

4-Hormonal ttt: e.g. O.C.P.: which suppress ovulation.

III- *Surgical measures*: Value
- → failed medical ttt
- → severe cases

Methods:

1- Dilatation of cervix. value
- → ttt of cervical stenosis → up to Hegar 6
- → lacerate paracervical nerve endings → up to Hegar 14

2- LUNA: laparoscopic uterine nerve ablation .

3- Presacral neuroectomy:

IV- *Acupuncture.*

2- Congestive Dysmenorrhea

Definition:

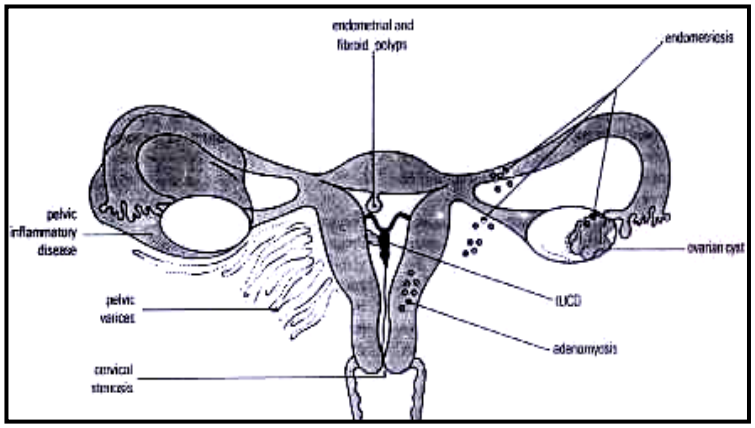
1- It's lower abdomen dull aching pain, referred to back.

2- Not accompanied by nausea and vomiting.

3- The pain starts few days (3-5 day) before menses and increase severity as menstruation approach and is relieved by the onset of menstrual flow.

4- It occurs in parous women.

5- Sometimes called secondary dysmenorrhea.



Causes: Pelvic congestion

1- Simple congestion → - constipation
→ - coitus interruptus.

2- Chronic inflammation

3- Displacement as RVF and prolapse .

4- Neoplasms e.g. fibroid.

5- Endometriosis: having special type of congestive dysmenorrheal:

a- type of pain: the pain ↑ instead of being ↓ by menstrual flow due to distension of ectopic endometrial glands .

b- grinding in character.

c- away from midline.

Treatment: treatment of the cause

+ glycerine icthyol pessary to decrease pelvic congestion.

N.B.: C/P. and investigation and ttt directed to the cause in congestive dysmenorrhea.

3) Membranous Dysmenorrhea

It is type of spasmodic dysmenorrhea but, .

1- Rare.
2- More severe pain.
3- Endometruim shed on the form of big membranous casts.
4- Of familial tendency.
5- Difficult treatment; may include repeated curettage

N.B.: D.D. of uterine casts:

1- Decidual casts → with villi → abortion
→ with no villi → disturbed ectopic pregnancy

2- Membranous dysmenorrhea.

3- Blood casts in severe uterine infection.

4) Ovarian Dysmenorrhea

1- It is type of congestive dysmenorrhea due to: ovarian congestion + thickened tunica albuginea of the ovary.
2- It occurs in one iliac fossa only.
3- The pain is reproduced by pressure on ovaries during bimanual examination.

N.B.: Ovulatory pain (Mittleschmerz pain):

It is mid cycle pain in one iliac fossa due to ↑ tension in mature follicle or rupture of follicle at ovulation:

It lasts only for 12hr.

PREMENSTRUAL TENSION SYNDROME (PTS)

Definition: Distressing physical, psychological and behavioural symptoms, not caused by organic disease, which regularly occur during the same phase of the menstrual (ovarian) cycle, and which significantly regress or disappear during the remainder of the cycle.

Causes : Many theories can be contributed :

1- Salt and water retention due to increase estrogen.
2- Progesterone deficiency.
3- Alteration of brain neurotransmitters → decrease dopamine with increase prolactin.
4- Decrease neurotransmitters (serotonin, catecholamines).
5- Vitamin deficiency (A, B6).
6- Reactive hypoglyacaemia.
7- Prostaglandins either (excess or deficiency).
8- Magnesium deficiency.
9- Allergic phenomenon.
10- Increase secretion of aldosterone.

C/P:

A: Anxiety.
B: Breast fullness and tenderness.
C: Cramps.
D: Depression.
E: Emotional instability.
F: Flatulence.
G: Gain of weight
H: Headache.
I: Irritability and insomnia.
J: Judgment disorders
L: Loneliness.

Treatment:

[A] Pharmacological therapy:

1- Premenstrual intake of diuretics.

2- Progesterone
3- Androgen
4- Bromocryptine.
} Anti estrogenic

5- Anti-histamincs in pre-menstrual period.

6- Vitamin B6 (as it's involved in synthesis of dopamine)

7- Spironlactone: which ↓ secretion of aldosterone.

8- Mild tranquilizers and psychotherapy for resistant cases.

9- Essential fatty acids: Evening primrose oil (EPO).

10- Calcium or magnesium supplementation.

[B] Non Pharmacological :

1- Exercise techniques (like yoga).

2- Acupuncture.

3- Stress reduction.

4- Herbalism.

5- Decrease smoking, caffeine, alcohol.

6- Diet: low fat, high protein with increases fibres.